Meteo Station — Power Subsystem (generated)

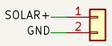
PV–12W –> [U6 INA219] –> CN3791 12V MPPT –> 1S LiPo –> [U7 INA219] –> MT3608 (5.0V) –> DevKit 5V pin.

D1 (1N5817): MT3608 5V –> DevKit 5V pin; isolates the boost so USB VBUS wins when flashing (no back–feed).

U6 INA219 0x40: high–side on the PV feed = harvest current + panel voltage.

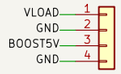
U7 INA219 0x41: high–side on the battery–>boost feed = load current + battery voltage (Vbus).

Both INA219 on I2C0 (VCC=3V3, 0.1ohm onboard shunt). GPIO2/ADC1_CH1 now a free spare; battery V comes from U7.

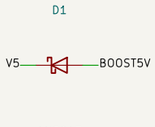


PV–12W 12V 1A

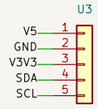
CN3791 12V MPPT



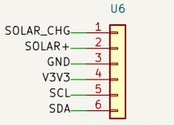
MT3608 –> 5.0V



1N5817



ESP32–H2–DevKitM–1



INA219 0x40 (PV)

1S LiPo 3.7V 10Ah

22uF

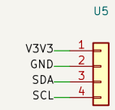
100uF

10uF

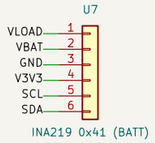
4.75k

4.75k

BMP388 0x76

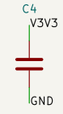


MLX90614 0x5A



INA219 0x41 (BATT)

100nF



10uF

100nF

Capacitor placement (physical):

C1 22uF – bulk at the MT3608 input (VLOAD, right at boost VIN, after the U7 shunt)

C2 100uF – bulk on the 5V rail, at the DevKit 5V pin

C3 10uF – beside C2 at the DevKit 5V pin

C4 10uF – bulk on the 3V3 rail, at the DevKit 3V3 pin

C5 100nF – across BMP388 VCC–GND, hard against the chip's pins

C6 100nF – across MLX90614 VCC–GND, hard against the chip's pins

Rule: a 100nF sits right at each chip; bulk caps go where current steps (boost out, radio, regulator).

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File: meteo_power.kicad_sch

Title: Meteo Station Power

Size: A3	Date:	**Rev: 1.0**
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